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MAGNETIC DISK DEVICE AND METHOD

Abstract

According to an embodiment, a controller corrects an instruction value of a VCM current by a first feedforward control based on a first detection value obtained by a first sensor. The controller executes a first operation in accordance with occurrence of a set event. In the first operation, the controller acquires a first waveform that is a waveform of a deviation amount of a detection position from a target position of a magnetic head, and acquires from the first waveform a first frequency band that is a frequency band whose amplitude is greater than a first threshold. The controller adjusts a coefficient of a transfer function of the first feedforward control to suppress the first detection value in a second frequency band different from the first frequency band and output a correction amount for a first instruction value based on the first detection value in the first frequency band.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-031640, filed on Mar. 1, 2024; the entire contents of which are incorporated herein by reference.

FIELD

[0002] Embodiments described herein relate generally to a magnetic disk device and a method.

BACKGROUND

[0003] In a magnetic disk device, a magnetic head is moved in a radial direction of a magnetic disk by a voice coil motor (VCM). In order to suppress the influence of disturbance on the positioning accuracy of the magnetic head, feedforward control based on various sensor values may be executed on an instruction value of the current that drives the VCM.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. 1 is a schematic diagram illustrating an example of a configuration of a magnetic disk device according to an embodiment;

[0005] FIG. 2 is a diagram illustrating an example of a configuration of the magnetic disk according to the embodiment;

[0006] FIG. 3 is a diagram illustrating an example of a control block in a seek operation, included in the magnetic disk device according to the embodiment;

[0007] FIG. 4 is a flowchart illustrating an example of operations of the magnetic disk device according to the embodiment;

[0008] FIG. 5 is a diagram illustrating an example of various waveforms acquired in the magnetic disk device according to the embodiment; and

[0009] FIG. 6 is a diagram illustrating an example of a method of adjusting a coefficient of a numerator and a coefficient of a denominator of an IIR type transfer function of an FF controller according to the embodiment.

DETAILED DESCRIPTION

[0010] In general, according to one embodiment, a magnetic disk device includes a magnetic disk, a magnetic head, an actuator arm, a motor, a drive circuit, a first sensor, and a controller. The magnetic head is configured to write and read data to and from the magnetic disk. The actuator arm has the magnetic head at a distal end. The motor is configured to move the actuator arm to move the magnetic head in a radial direction of the magnetic disk. The drive circuit is configured to generate a first current that drives the motor by an amount according to an instruction value and supply the generated first current to the motor. The first sensor is configured to detect a predetermined physical quantity. The controller is configured to: detect a position of the magnetic head; generate a first instruction value so as to reduce a deviation amount of a detection position of the magnetic head from a target position of the magnetic head; correct the first instruction value by a first feedforward control based on a first detection value obtained by the first sensor; input a second instruction value, which is a value obtained by correcting the first instruction value, to the drive circuit as the instruction value; and execute a first operation in accordance with occurrence of a set event. In the first operation, the controller is configured to adjust a coefficient of a transfer

function of the first feedforward control by the following operation. That is, the controller is configured to acquire a first waveform that is a waveform of the deviation amount, and acquire from the first waveform a first frequency band that is a frequency band whose amplitude is greater than a first threshold. The controller is configured to adjust the coefficient of the transfer function of the first feedforward control so as to suppress the first detection value in a second frequency band different from the first frequency band and output a correction amount for the first instruction value based on the first detection value in the first frequency band.

[0011] Exemplary embodiments of a magnetic disk device and a method will be explained below in detail with reference to the accompanying drawings. The present invention is not limited to the following embodiments.

Embodiments

[0012] FIG. 1 is a schematic diagram illustrating an example of a configuration of a magnetic disk device 1 according to an embodiment.

[0013] The magnetic disk device 1 is connected to a host 2. The magnetic disk device 1 can receive an access command such as a write command or a read command from the host 2.

[0014] The magnetic disk device 1 includes a magnetic disk 11 having a magnetic layer formed on a surface thereof. The magnetic disk device 1 writes data to the magnetic disk 11 or reads data from the magnetic disk 11 in response to the access command.

[0015] Data is written and read via a magnetic head 22. The magnetic disk device 1 includes, in addition to the magnetic disk 11, a spindle motor 12, a ramp 13, an actuator arm 15, a voice coil motor (VCM) 16, a first vibration sensor 17a, a second vibration sensor 17b, a motor driver integrated circuit (IC) 21, a magnetic head 22, a hard disk controller (HDC) 23, a head IC 24, a read/write channel (RWC) 25, a processor 26, RAM 27, flash read only memory (FROM) 28, and buffer memory 29.

[0016] The magnetic disk 11 is rotated at a predetermined rotation speed by the spindle motor 12 attached coaxially. Rotation of the spindle motor 12 is driven by the motor driver IC 21.

[0017] The motor driver IC 21 drives the spindle motor 12 and the VCM 16 under the control of the processor 26. In particular, regarding the VCM 16, the motor driver IC 21 receives, from the processor 26, an input of an instruction value regarding the current to be supplied to the VCM 16, and generates a current of an amount corresponding to the instruction value. The motor driver IC 21 then supplies the generated current to the VCM 16. The current supplied to the VCM 16 is referred to as a VCM current. An instruction value related to the VCM current input from the processor 26 to the motor driver IC 21 is referred to as a VCM current instruction value.

[0018] The magnetic head 22 uses its own elements of a write element 22w and a read element 22r to write and read data to and from the magnetic disk 11. The magnetic head 22 is attached to a distal end of the actuator arm 15. The VCM 16 moves the actuator arm to move the magnetic head 22 in the radial direction of the magnetic disk 11. Note that one or both of the write element 22w and the read element 22r on the magnetic head 22 may be provided in plurality for a single magnetic head 22.

[0019] The VCM 16 is an example of a motor. The motor driver IC 21 is an example of a drive circuit.

[0020] For example, the magnetic head 22 is moved onto the ramp 13 during the time such as when the rotation of the magnetic disk 11 is stopped. The ramp 13 is configured to hold the magnetic head 22 at a position separated from the magnetic disk 11.

[0021] The head IC 24 amplifies and outputs a signal read from the magnetic disk 11 by the magnetic head 22 during the read operation, and supplies the amplified signal to the RWC 25. In addition, the head IC 24 amplifies, during the write operation, a signal corresponding to write target data supplied from the RWC 25 and supplies the amplified signal to the magnetic head 22.

[0022] The HDC 23 performs control of transmission and reception of data with the host 2 via the I/F bus, control of the buffer memory 29, and the like.

[0023] The buffer memory **29** is used as a buffer for data transmitted to and received from the host **2**. For example, the buffer memory **29** is used to temporarily store write target data or data read from the magnetic disk **11**.

[0024] The buffer memory **29** is formed with volatile memory capable of high-speed operation. The type of the memory constituting the buffer memory **29** is not limited to a specific type. The buffer memory **29** can be constituted with memory such as Dynamic Random Access Memory (DRAM) and Static Random Access Memory (SRAM), for example. The buffer memory **29** may be constituted with any types of non-volatile memory.

[0025] The RWC **25** performs modulation such as error correction coding on write target data supplied from the HDC **23**, and supplies the modulated data to the head IC **24**. In addition, the RWC **25** performs demodulation including error correction on the signal read from the magnetic disk **11** and supplied from the head IC **24**, and then outputs the demodulated signal to the HDC **23** as digital data.

[0026] An example of the processor **26** is a central processing unit (CPU). The processor **26** is connected to the first vibration sensor **17a**, the second vibration sensor **17b**, the RAM **27**, the FROM **28**, and the buffer memory **29**.

[0027] The FROM **28** is nonvolatile memory. The FROM **28** stores firmware (program data), various operation parameters, and the like. The operation parameter may include, for example, thresholds Th1, Th2, and Th3 described below. Note that the firmware or various operation parameters may be stored in a predetermined area of the magnetic disk **11**.

[0028] The RAM **27** includes, for example, DRAM, SRAM, or a combination thereof. The RAM **27** is used as operation memory by the processor **26**. The RAM **27** is used as an area in which firmware is loaded and an area in which various types of operation parameters are temporarily stored.

[0029] The processor **26** performs overall control of the magnetic disk device **1** according to the firmware stored in the FROM **28** or the magnetic disk **11**. For example, the processor **26** loads the firmware from the FROM **28** or the magnetic disk **11** into the RAM **27**, and executes control of the motor driver IC **21**, the head IC **24**, the RWC **25**, the HDC **23**, and the like according to the loaded firmware.

[0030] The first vibration sensor **17a** and the second vibration sensor **17b** can detect the amount of vibration. The amount of vibration detected by the vibration sensor **17** is displacement, speed, acceleration, or any other physical quantity. In order to suppress the influence of vibration received by the magnetic disk device **1** from the outside or vibration generated in the magnetic disk device **1** on the positioning control accuracy, the processor **26** performs feedforward control on the VCM current instruction value based on the detection values obtained by the first vibration sensor **17a** and the second vibration sensor **17b**. Details of the feedforward control on the VCM current instruction value will be described below.

[0031] The configuration including the HDC **23**, the RWC **25**, and the processor **26** can also be regarded as a controller **30** that controls operations of the magnetic disk device **1**. In addition to these components, the controller **30** may include other components (For example, the RAM **27**, the FROM **28**, the buffer memory **29**, and the like). The controller **30** may be formed as one system-on-a-chip (SoC) or may be formed with two or more chips.

[0032] Part or all of the functions of the processor **26** may be implemented by a hardware circuit such as a field-programmable gate array (FPGA) or an application specific integrated circuit (ASIC).

[0033] The number of the magnetic disks **11** included in the magnetic disk device **1** is not limited to one. The magnetic disk device **1** may include actuator arms **15** and magnetic heads **22** in a number corresponding to the number of magnetic disks **11**. In a case where the magnetic disk device **1** includes a plurality of magnetic heads **22**, the plurality of magnetic heads **22** may be integrally moved, or the plurality of magnetic heads **22** may constitute a plurality of groups each being

independently movable.

[0034] FIG. 2 is a diagram illustrating an example of a configuration of the magnetic disk **11** according to the embodiment. Servo data used for positioning the magnetic head **22** is written to a magnetic layer formed on a surface of the magnetic disk **11** by using a servo writer or self-servo write (SSW), for example.

[0035] FIG. 2 illustrates a servo region **41** radially arranged as an example of arrangement of a servo region in which the servo data is written. In the circumferential direction, a space between the two servo regions **41** is defined as a data region **42** in which data can be written. The magnetic disk **11** includes a plurality of concentric tracks (for example, tracks **50** in the drawing) set in the radial direction. The region delimited by the servo region **41** on the track **50** is also referred to as a servo sector.

[0036] The servo data includes a servo mark, Gray code, a burst pattern, and a post code. The servo mark indicates the start of the servo data. The Gray code includes an ID for identifying each track **50** provided in the magnetic disk **11**, that is, a track number, and an ID for identifying each servo sector (that is, the servo region **41**) on the track **50**, namely, a servo sector number. The burst pattern is data used to detect the amount of position shift of the track indicated by the track number included in the Gray code from the center. The track number included in the Gray code is given as an integer value, for example. Demodulation of the burst pattern makes it possible to obtain an offset amount of decimal places based on the position indicated by the track number. That is, by demodulating the burst pattern, the current position of the magnetic head **22** in the radial direction is obtained. The post code is data for correcting position shift of the shape of the track **50** defined by the Gray code and the burst pattern from the ideal shape of the track **50**.

[0037] When writing data to the magnetic disk **11** or reading data from the magnetic disk **11**, the controller **30** executes positioning control of the magnetic head **22** based on the servo data read by the magnetic head **22** from the servo region **41**. In the positioning control, the controller **30** generates the VCM current instruction value so as to reduce the deviation amount (hereinafter, it is described as a position deviation amount) of the detected position from the target position of the magnetic head **22**.

[0038] The positioning control includes a control of a seek operation and a control of a tracking operation. The seek operation is an operation of moving the magnetic head **22** in the radial direction toward the target track. The tracking operation is an operation of maintaining the magnetic head **22** on a target track.

[0039] In the seek operation, the controller **30** executes a seek settling operation. The seek settling operation is an operation of determining whether to start an access (that is, write or read) to the magnetic disk **11** based on the position deviation amount. In one example, the seek settling operation includes determination of whether the position deviation amount has continued to fall within a set range for a predetermined time. When it is confirmed that the position deviation amount has continued to fall within the set range for a predetermined time, the seek settling operation (and seek operation) are completed, the tracking operation is started, and write or read to/from the magnetic disk **11** is started while the tracking operation is under control.

[0040] In the embodiment, the controller **30** performs feedforward control on the VCM current instruction value in the seek operation. The controller **30** performs feedforward control on the VCM current instruction value to suppress the influence of disturbance such as vibration on the positioning accuracy in the seek operation.

[0041] Note that the feedforward control on the VCM current instruction value can be performed based on a detection value obtained by an optionally selected sensor in addition to or instead of the detection value obtained by the first vibration sensor **17a** and the detection value obtained by the second vibration sensor **17b** described above. Here, as an example, it is assumed that the feedforward control on the VCM current instruction value is executed based on the detection value obtained by the first vibration sensor **17a**, the detection value obtained by the second vibration

sensor **17b**, and the detection value of the VCM current supplied to the VCM **16**.

[0042] Hereinafter, the detection value obtained by the first vibration sensor **17a** is referred to as a first vibration detection value. The detection value obtained by the second vibration sensor **17b** is referred to as a second vibration detection value. The detection value of the VCM current supplied to the VCM **16** is referred to as a VCM current detection value.

[0043] FIG. **3** is a diagram illustrating an example of a control block in a seek operation, included in the magnetic disk device **1** according to the embodiment.

[0044] The magnetic disk device **1** includes a target speed table **101**, a VCM controller **102**, a VCM driver **103**, a VCM current detector **104**, a feed forward (FF) controller adjustment mechanism **105**, a first FF controller **106**, a second FF controller **107**, a third FF controller **108**, a head speed estimator **109**, a settling determination mechanism **110**, and four adders **111** to **114**.

[0045] The target speed table **101** is stored in the FROM **28**, for example, and read by the processor **26**. The VCM controller **102**, the FF controller adjustment mechanism **105**, the first FF controller **106**, the second FF controller **107**, the third FF controller **108**, the head speed estimator **109**, the settling determination mechanism **110**, and the four adders **111** to **114** are implemented by the processor **26**. The VCM driver **103** and the VCM current detector **104** are included in the motor driver IC **21**.

[0046] In the description of the present specification, addition includes not only calculation of addition of a positive value but also calculation of addition of a negative value, that is, subtraction. That is, some or all of the four adders **111** to **114** can also perform subtraction.

[0047] The magnetic disk device **1** controls the control target Ctgt by these functional components. The control target Ctgt includes the VCM **16**, the actuator arm **15**, the magnetic head **22**, the head IC **24**, and the RWC **25**. The processor **26** detects the current position of the magnetic head **22** in the radial direction based on the servo data read from the servo sector, and acquires the detected position.

[0048] In addition, the processor **26** performs a predetermined calculation to acquire a target position Ptgt, which is a target position in the radial direction. The adder **111** performs calculation of subtracting the detection position from the target position Ptgt to acquire the position deviation amount in the radial direction.

[0049] The processor **26** acquires the target speed in the radial direction of the magnetic head **22** based on the position deviation amount and the target speed table **101**.

[0050] The target speed table **101** includes information that defines a relationship between a seek distance and a target speed of the magnetic head **22** in the radial direction. The seek distance is a distance in the radial direction from the current position of the magnetic head **22** to the target track. In the target speed table **101**, for example, the relationship between the seek distance and the target speed of the magnetic head **22** is defined so that the longer the seek distance, the higher the target speed of the magnetic head **22** will be. The processor **26** searches the target speed table **101** using the position deviation amount as the seek distance, thereby acquiring the target speed of the magnetic head **22** corresponding to the position deviation amount.

[0051] The target speed table **101** may represent information in the form of a table or may be in the form of a mathematical expression. The processor **26** may further perform fine adjustment on the target speed obtained based on the target speed table **101**.

[0052] The speed of the magnetic head **22** and the speed of the VCM **16** can be easily converted with each other. Accordingly, the target speed may be expressed as the motor speed of the VCM **16**. Hereinafter, the speed (including the target speed and the like) will be described as the speed of the magnetic head **22** in the radial direction. Furthermore, the speed of the magnetic head **22** is to be represented as the speed of the magnetic head **22** in the radial direction.

[0053] The head speed estimator **109** estimates the speed of the magnetic head **22** based on the position deviation amount and the VCM current instruction value.

[0054] The adder **112** performs calculation of subtracting the estimated value of the speed of the

magnetic head **22** obtained by the head speed estimator **109** from the target speed, thereby acquiring the speed deviation amount.

[0055] The VCM controller **102** calculates a VCM current instruction value based on the speed deviation amount. The VCM controller **102** calculates the VCM current instruction value so that the speed deviation amount decreases, in other words, the speed of the magnetic head **22** in the radial direction approaches the target speed. In one example, the VCM controller **102** calculates the VCM current instruction value such that the greater the absolute value of the speed deviation amount, the greater the VCM current instruction value will be.

[0056] The adder **113** corrects the VCM current instruction value by calculation of adding a correction amount by the feedforward control to the VCM current instruction value obtained by the VCM controller **102**. The VCM current instruction value before correction by the adder **113** is referred to as a first VCM current instruction value. The VCM current instruction value after correction by the adder **113** is referred to as a second VCM current instruction value.

[0057] The VCM current instruction value used by the head speed estimator **109** to estimate the speed of the magnetic head **22** is the first VCM current instruction value.

[0058] The VCM driver **103** generates the VCM current of an amount corresponding to the second VCM current instruction value. The VCM current generated by the VCM driver **103** is supplied to the VCM **16** among the control target Ctgt.

[0059] The VCM current detector **104** detects the amount of the VCM current generated by the VCM driver **103** and supplied to the VCM **16**. The amount of the VCM current detected by the VCM current detector **104** is referred to as a VCM current detection value.

[0060] The first FF controller **106** calculates a correction amount to be used in the feedforward control based on the detection value obtained by the first vibration sensor **17a**. The second FF controller **107** calculates a correction amount to be used in feedforward control based on the detection value obtained by the second vibration sensor **17b**. The third FF controller **108** calculates a correction amount to be used in the feedforward control based on the VCM current detection value.

[0061] The adder **114** adds the correction amounts acquired by the first FF controller **106**, the second FF controller **107**, and the third FF controller **108**. The sum of the correction amounts obtained by the adder **114** is input to the adder **113**. The adder **113** adds the sum of the correction amounts input from the adder **114** to the first VCM current instruction value, thereby acquiring the second VCM current instruction value.

[0062] The first vibration sensor **17a**, the second vibration sensor **17b**, and the VCM current detector **104** each are an example of a second sensor. One of the first vibration sensor **17a**, the second vibration sensor **17b**, and the VCM current detector **104** is an example of a first sensor that detects a predetermined physical quantity. The physical quantity detected by the first vibration sensor **17a** and the second vibration sensor **17b** is vibration. The physical quantity detected by the VCM current detector **104** is the VCM current.

[0063] The correction amount by each of the first FF controller **106**, the second FF controller **107**, and the third FF controller **108** is obtained by executing calculation using an infinite impulse response (IIR) filter. That is, each of the first FF controller **106**, the second FF controller **107**, and the third FF controller **108** has an IIR type transfer function.

[0064] Having internal feedback, the IIR filter is capable of complicated control with a simpler structure as compared with a finite impulse response (FIR) filter. If control equivalent to that of the IIR filter is to be performed by the FIR filter, it is inevitable to increase the order of the FIR filter, which would increase the calculation amount of the FIR filter. Here, the calculation of the correction amount by each of the first FF controller **106**, the second FF controller **107**, and the third FF controller **108** is executed using an IIR filter, making it possible to perform feedforward control with high accuracy with a low calculation amount.

[0065] The FF controller adjustment mechanism **105** individually adjusts the coefficient of the

transfer function of the IIR filter used in the first FF controller **106**, the coefficient of the transfer function of the IIR filter used in the second FF controller **107**, and the coefficient of the transfer function of the IIR filter used in the third FF controller **108**. The FF controller adjustment mechanism **105** adjusts both the numerator coefficient and the denominator coefficient for the transfer function of each IIR filter.

[0066] The settling determination mechanism **110** determines whether a seek settling operation is being executed based on the position deviation amount. While the seek settling operation is being executed, a signal indicating that the seek settling operation is being executed is input to the FF controller adjustment mechanism **105**. A signal indicating whether the seek settling operation is being executed is referred to as a settling flag.

[0067] The transition of the position deviation amount during the period of the seek settling operation is related to the accuracy of positioning control. In a case where the accuracy of the positioning control is deteriorated due to disturbance (here, vibration or fluctuation of the VCM current) or the like, the transition of the position deviation amount during the period of the seek settling operation indicates an increased amplitude of the position deviation amount or a longer time required for the seek settling operation compared with a case where the accuracy of the positioning control is good. That is, the transition of the position deviation amount during the period of the seek settling operation is considered to reflect the influence of disturbance.

Accordingly, the settling determination mechanism **110** adjusts the coefficients of the transfer functions of the individual IIR filters of the first FF controller **106**, the second FF controller **107**, and the third FF controller **108** based on the transition of the position deviation amount in the period of the seek settling operation.

[0068] Incidentally, the FF controller adjustment mechanism **105** receives input of not only the settling flag, but also the position deviation amount, the detection value obtained by the first vibration sensor **17a**, the detection value obtained by the second vibration sensor **17b**, and the VCM current detection value. Based on the input information, the FF controller adjustment mechanism **105** adjusts the coefficients of the transfer functions of the individual IIR filters of the first FF controller **106**, the second FF controller **107**, and the third FF controller **108**.

[0069] The timing of adjustment of the coefficient of the transfer function of each IIR filter is not limited to a specific timing. The controller **30** may periodically adjust the coefficients of the transfer function of each IIR filter, or may adjust the coefficients of the transfer function of each IIR filter in accordance with occurrence of a specific event.

[0070] Here, as an example, the coefficient of the transfer function of each IIR filter is supposed to be adjusted when the positioning accuracy no longer satisfies a predetermined criterion. More specifically, the controller **30** monitors Input/Output Operations per Second (IOPS). IOPS is the number of processes per unit time executed by the magnetic disk device **1**, and is one of indices indicating the performance of the magnetic disk device **1**. At a timing at which IOPS falls below a predetermined threshold (referred to as a threshold Th1), the FF controller adjustment mechanism **105** adjusts the coefficient of the transfer function of each IIR filter. It is conceivable that the positioning accuracy deterioration leads to an increase of seek errors and reduction of IOPS. Therefore, when IOPS falls below the threshold Th1 preset value as a reference value, the controller **30** adjusts the coefficient of the transfer function of each IIR filter for recovery of the positioning accuracy.

[0071] FIG. **4** is a flowchart illustrating an example of operations of the magnetic disk device **1** according to the embodiment.

[0072] The controller **30** determines whether IOPS is smaller than the threshold Th1 (S**101**).

[0073] When IOPS is not smaller than the threshold Th1 (S**101**: No), the control proceeds to S**114** described below.

[0074] When IOPS is smaller than the threshold Th1 (S**101**: Yes), the controller **30** adjusts the coefficient of the transfer function of each IIR filter by the operations of S**102** to S**113**. The

operations of **S102** to **S113** are an example of the first operation.

[0075] In the first operation, the controller **30** first determines whether an execution timing of the seek operation has arrived (**S102**). When the execution timing of the seek operation has not arrived (**S102**: No), the controller **30** executes the process of **S102** again.

[0076] When the execution timing of the seek operation has arrived (**S102**: Yes), the FF controller adjustment mechanism **105** turns off the function of the feedforward control by the first FF controller **106**, the second FF controller **107**, and the third FF controller **108** (**S103**). In **S103**, the FF controller adjustment mechanism **105** causes the first FF controller **106**, the second FF controller **107**, and the third FF controller **108** to each output 0 as the correction amount, for example. This causes the first FF controller **106**, the second FF controller **107**, and the third FF controller **108** to be in a state of not executing feedforward control, so as to start positioning control under the influence of disturbance such as vibration and fluctuation of the VCM current.

[0077] The FF controller adjustment mechanism **105** starts acquisitions of individual values, namely, the first vibration detection value, the second vibration detection value, the VCM current detection value, and the position deviation amount (**S104**). The controller **30** starts a seek operation (**S105**).

[0078] Incidentally, the detection position is acquired from each servo sector in synchronization with the rotation angle of the magnetic disk **11**. The controller **30** constitutes a sampled-data control system that determines an input to a control target (for example, the VCM **16**) at regular time intervals. The VCM **16** is driven in 1/N cycle (N is an integer of 2 or more) of the acquisition cycle of the detection position, that is, can undergo multi-rate control. However, the acquisition timing of the first vibration detection value, the second vibration detection value, the VCM current detection value, and the position deviation amount is set as the same as the acquisition timing of the detection position. During the period of seek operation, time transitions of the first vibration detection value, the second vibration detection value, the VCM current detection value, and the position deviation amount are acquired as waveforms.

[0079] When the seek operation is completed (**S106**), the FF controller adjustment mechanism **105** ends acquisitions of the first vibration detection value, the second vibration detection value, the VCM current detection value, and the position deviation amount (**S107**).

[0080] The FF controller adjustment mechanism **105** acquires the waveform during the seek settling operation from the waveform of the position deviation amount acquired in the period of the seek operation (**S108**).

[0081] During the period of the seek settling operation among the seek operation, the settling determination mechanism **110** maintains the settling flag in an on state. Based on the settling flag, the FF controller adjustment mechanism **105** specifies and acquires the waveform during the seek settling operation from the waveform of the position deviation amount. The waveform of the position deviation amount during the seek settling operation acquired by the process of **S108** is referred to as a settling position waveform.

[0082] FIG. 5 is a diagram illustrating an example of various waveforms acquired in the magnetic disk device **1** according to the embodiment. This drawing illustrates waveforms of the position deviation amount, the VCM current detection value, the first vibration detection value, the second vibration detection value, and the settling flag from the start of the seek operation to the completion of the seek operation.

[0083] When the seek operation is started, an acceleration period in which the magnetic head **22** is greatly accelerated toward the target track starts. When the magnetic head **22** approaches the target track, a deceleration period in which the magnetic head **22** is greatly decelerated starts. When the position deviation amount becomes a predetermined value or less after the deceleration period, the seek settling operation is executed. By maintaining the settling flag on, the FF controller adjustment mechanism **105** is notified that a seek settling operation is being performed. Regarding the position deviation amount, the FF controller adjustment mechanism **105** acquires the waveform

during the seek settling operation.

[0084] Regarding the VCM current detection value, the first vibration detection value, and the second vibration detection value, waveforms in a period from the disclosure of the seek operation to the completion of the seek operation (in other words, the seek settling operation) are acquired.

[0085] The settling position waveform is an example of the first waveform. The waveform of the first vibration detection value, the waveform of the second vibration detection value, and the waveform of the VCM current detection value each are an example of a second waveform.

[0086] Now, description will continue by returning to FIG. 4. Following the process of S108, the FF controller adjustment mechanism 105 performs frequency analysis on the waveform of the first vibration detection value, the waveform of the second vibration detection value, the waveform of the VCM current detection value, and the settling position waveform (S109).

[0087] The frequency analysis is an analysis using a discrete Fourier transform, for example. The FF controller adjustment mechanism 105 performs frequency analysis to acquire characteristics for each waveform, specifically, a first characteristic indicating a relationship between a frequency and an amplitude (expressed as a gain) and a second characteristic indicating a relationship between a frequency and a phase.

[0088] Based on the first characteristic related to the settling position waveform, the FF controller adjustment mechanism 105 specifies a frequency band in which the gain is greater than a threshold Th2 in the settling position waveform (S110).

[0089] The threshold Th2 is a preset reference value for determining whether correction by feedforward control is necessary. That is, a frequency band in which the gain is greater than the threshold Th2 is regarded as a frequency band that needs correction for the VCM current instruction value by feedforward control; a frequency band in which the gain is not greater than the threshold Th2 is regarded as a frequency band that needs no correction for the VCM current instruction value by feedforward control. S110 specifies a frequency band that needs a correction by feedforward control. The frequency band acquired by the process of S110 is referred to as a target frequency band.

[0090] The target frequency band is an example of the first frequency band. The frequency band other than the target frequency band is an example of the second frequency band.

[0091] The FF controller adjustment mechanism 105 acquires coherence in the target frequency band between: waveforms of the first vibration detection value, the second vibration detection value, and the VCM current detection value; and the settling position waveform (S111).

[0092] Subsequently, the FF controller adjustment mechanism 105 specifies a physical quantity whose coherence in the target frequency band is greater than a threshold Th3 among the vibration detected by the first vibration sensor 17a, the vibration detected by the second vibration sensor 17b, and the VCM current detected by the VCM current detector 104 (S112).

[0093] S112 specifies a disturbance that has caused deterioration of positioning control. The threshold Th3 is a present reference value for determining whether a physical quantity to be detected by each sensor is a disturbance that has caused deterioration of positioning control. A physical quantity whose coherence is greater than the threshold Th3 is considered as a disturbance that has caused deterioration of the positioning control, while a physical quantity whose coherence is not greater than the threshold Th3 is not considered as a disturbance that has caused the deterioration of the positioning control. The physical quantity specified by the process of S112 is referred to as a target physical quantity.

[0094] The FF controller adjustment mechanism 105 adjusts a coefficient of a transfer function of an FF controller that performs feedforward control based on a target physical quantity, among the first FF controller 106, the second FF controller 107, and the third FF controller 108 (S113). S113 adjusts both the coefficient of the numerator and the coefficient of the denominator of the IIR type transfer function of the FF controller that performs the feedforward control based on the target physical quantity.

[0095] FIG. 6 is a diagram illustrating an example of a method of adjusting a coefficient of a numerator and a coefficient of a denominator of an IIR type transfer function of an FF controller according to the embodiment. (A) of FIG. 6 is a graph illustrating the first characteristic, and (B) of FIG. 6 is a graph illustrating the second characteristic. The graph of (A) of FIG. 6 and the graph of (B) of FIG. 6 individually illustrate a frequency characteristic of the target physical quantity and a target frequency characteristic (hereinafter, target frequency characteristic) of the FF controller that performs feedforward control based on the target physical quantity.

[0096] As illustrated in (A) and (B) of FIG. 6, the FF controller adjustment mechanism **105** designs the target frequency characteristic so as to suppress the target physical quantity in a frequency band other than the target frequency band and pass the target physical quantity in the target frequency band.

[0097] The FF controller adjustment mechanism **105** calculates a coefficient of a numerator and a coefficient of a denominator of the transfer function of the FF controller so that the frequency characteristic of the transfer function of the FF controller matches or approaches the target frequency characteristic. A calculation algorithm used to calculate the coefficient of the numerator and the coefficient of the denominator of the transfer function for achieving the target frequency characteristic is not limited to a specific algorithm. The FF controller adjustment mechanism **105** calculates the coefficient of the numerator and the coefficient of the denominator of the transfer function of the FF controller based on any known calculation algorithm.

[0098] Now, description will continue by returning to FIG. 4 again. When the seek operation is executed next after the process of **S113**, the FF controller adjustment mechanism **105** turns on the functions of the feedforward control by the first FF controller **106**, the second FF controller **107**, and the third FF controller **108** (**S114**). The control proceeds to **S101**.

[0099] In the example described above, the magnetic disk device **1** includes three sensors as sensors for detecting a physical quantity that can be a disturbance, namely, the first vibration sensor **17a**, the second vibration sensor **17b**, and the VCM current detector **104**. The number of sensors for detecting a physical quantity that may be a disturbance, included in the magnetic disk device, **1** is not limited to a specific number as long as the number is one or more.

[0100] In addition, the number of FF controllers included in the controller **30** is not limited to three. The controller **30** includes FF controllers as many as the number of sensors for detecting a physical quantity that can be a disturbance, and each FF controller executes feedforward control based on a physical quantity detected by a mutually different sensor.

[0101] In addition, as illustrated in **S112** and **S113** of FIG. 4, the controller **30** specifies an FF controller as a transfer function coefficient adjustment target among the three FF controllers (first FF controller **106**, second FF controller **107**, and third FF controller **108**) based on the coherence in the target frequency band. The method of specifying the FF controller as a transfer function coefficient adjustment target is not limited thereto. The controller **30** may set all the three FF controllers as transfer function coefficient adjustment target.

[0102] As described above, according to the embodiment, the controller **30** generates the first VCM current instruction value by the target speed table **101** and the VCM controller **102** so as to reduce the position deviation amount. The controller **30** corrects the first VCM current instruction value by feedforward control based on the physical quantity detected by the sensor, and inputs the second VCM current instruction value, which is the corrected first VCM current instruction value, to the motor driver IC **21**. The motor driver IC **21** causes the VCM driver **103** to generate a VCM current of an amount corresponding to the second VCM current instruction value, and supplies the generated VCM current to the VCM **16**. The controller **30** executes the first operation (refer to **S101** to **S113** in FIG. 4, for example) in accordance with occurrence of a set event. In the first operation, the controller **30** acquires the waveform of the position deviation amount, and acquires a frequency band in which the gain in the waveform of the position deviation amount is greater than the threshold **Th2** as the target frequency band. Subsequently, the controller **30** adjusts the

coefficient of the transfer function of the feedforward control so as to suppress the detection value obtained by the sensor in the frequency band other than the target frequency band and output the correction amount for the first VCM current instruction value based on the target frequency band. [0103] Since the controller **30** can autonomously adjust the coefficient of the transfer function of the feedforward control, it is possible to improve the positioning accuracy.

[0104] In addition, according to the embodiment, the magnetic disk device **1** includes a plurality of sensors that detect physical quantities, and the controller **30** corrects the first VCM current instruction value by a plurality of feedforward controls based on the physical quantities detected by different sensors. In the first operation, the controller **30** selects one or more feedforward controls among the plurality of feedforward controls as a transfer function coefficient adjustment target.

[0105] This makes it possible to improve the positioning accuracy.

[0106] In addition, according to the embodiment, the controller **30** acquires a plurality of waveforms from a plurality of sensors. The controller **30** acquires coherence of each of the plurality of waveforms acquired from the plurality of sensors and the waveform of the position deviation amount, in the target frequency band.

[0107] Subsequently, a feedforward control based on a physical quantity whose coherence is greater than the threshold Th_3 among the plurality of feedforward controls is specified, and a coefficient of a transfer function of the specified feedforward control is adjusted (refer to **S112** and **S113** in FIG. 4, for example).

[0108] That is, the physical quantity to be the disturbance causing the deterioration of the positioning control, among the plurality of physical quantities detected by the plurality of sensors, is specified, and the coefficient of the transfer function of the feedforward control based on the specified physical quantity is adjusted. This makes it possible to improve the positioning accuracy.

[0109] According to the embodiment, the controller **30** acquires the waveforms of the plurality of sensors during the seek operation, and acquires the waveform of the position deviation amount during the seek settling operation.

[0110] This makes it possible to improve the positioning accuracy in the seek operation, leading to improvement of the performance of the magnetic disk device **1**.

[0111] Note that the first operation may be executed in the tracking operation in addition to the seek operation or instead of the seek operation. For example, the controller **30** acquires waveforms of the plurality of sensors and the waveform of the position deviation amount during a part or all of the period of the tracking operation. The controller **30** may use these acquired waveforms to specify a target frequency band, specify feedforward control of a transfer function coefficient adjustment target based on coherence in the target frequency band, etc.

[0112] In addition, according to the embodiment, the controller **30** monitors the number of processes per unit time of the magnetic disk device **1**, and executes the first operation at a timing when the number of processes per unit time becomes smaller than the threshold Th_1 (refer to **S101** in FIG. 4, for example).

[0113] Therefore, when the performance is deteriorated, the performance can be recovered by adjusting the coefficient of the transfer function.

[0114] Further, according to the embodiment, each FF controller has an IIR type transfer function, and the controller **30** adjusts a coefficient included in a numerator of the transfer function and a coefficient included in a denominator of the transfer function.

[0115] The IIR filter can perform complicated control with a simpler structure as compared with the FIR filter. Since each FF controller performs correction by calculation using the IIR filter, it is possible to perform feedforward control with high accuracy with a low calculation amount. In addition, the controller **30** adjusts the coefficient included in the numerator of the transfer function and the coefficient included in the denominator of the transfer function, making it possible to allow the transfer function of the FF controller to have a frequency characteristic as close as possible to the target frequency characteristic. That is, the target frequency characteristic can be achieved with

a simple method and high accuracy.

[0116] The transfer function of each FF controller need not be the IIR type transfer function. The transfer functions of some or all of the FF controllers may be FIR type transfer functions.

[0117] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

Claims

1. A magnetic disk device comprising: a magnetic disk; a magnetic head configured to write and read data to and from the magnetic disk; an actuator arm having the magnetic head at a distal end; a motor configured to move the actuator arm to move the magnetic head in a radial direction of the magnetic disk; a drive circuit configured to generate a first current that drives the motor by an amount according to an instruction value and supply the generated first current to the motor; a first sensor configured to detect a predetermined physical quantity; and a controller configured to: detect a position of the magnetic head; generate a first instruction value so as to reduce a deviation amount of a detection position of the magnetic head from a target position of the magnetic head; correct the first instruction value by a first feedforward control based on a first detection value obtained by the first sensor; input a second instruction value, which is a value obtained by correcting the first instruction value, to the drive circuit as the instruction value; and execute a first operation in accordance with occurrence of a set event, the first operation being an operation of acquiring a first waveform that is a waveform of the deviation amount, acquiring from the first waveform a first frequency band that is a frequency band whose amplitude is greater than a first threshold, and adjusting a coefficient of a transfer function of the first feedforward control so as to suppress the first detection value in a second frequency band different from the first frequency band and output a correction amount for the first instruction value based on the first detection value in the first frequency band.

2. The magnetic disk device according to claim 1, further comprising: a plurality of second sensors each of which detects a predetermined physical quantity, the plurality of second sensors including the first sensor, wherein the controller is configured to correct the first instruction value by a plurality of second feedforward controls, the plurality of second feedforward controls including the first feedforward control, and each of the plurality of second feedforward controls being a control of generating a correction amount for the first instruction value based on a second detection value obtained by a mutually different one of the plurality of second sensors, and the controller is configured to adjust, in the first operation, a coefficient of a transfer function of each of one or more second feedforward controls among the plurality of second feedforward controls so as to suppress the second detection value in the second frequency band and generate a correction amount for the first instruction value based on the second detection value in the first frequency band.

3. The magnetic disk device according to claim 2, wherein in the first operation, the controller is configured to: acquire a plurality of second waveforms, the plurality of second waveforms being waveforms of second detection values respectively obtained from the plurality of second sensors; acquire coherence in the first frequency band between each of the plurality of second waveforms and the first waveform; specify, among the second detection values and the second waveforms, a second feedforward control based on a second detection value constituting a second waveform whose coherence is greater than a second threshold among the plurality of second feedforward controls; and adjust a coefficient of a transfer function of the specified second feedforward control.

4. The magnetic disk device according to claim 3, wherein each of the plurality of second waveforms includes a waveform during a seek operation, and the first waveform includes a waveform during a seek settling operation.
5. The magnetic disk device according to claim 1, wherein the transfer function includes an infinite impulse response (IIR) type transfer function in which coefficients are included in a numerator and a denominator, and the controller is configured to adjust, in the first operation, the coefficient included in the numerator of the transfer function and the coefficient included in the denominator of the transfer function.
6. The magnetic disk device according to claim 2, wherein the transfer function of each of the plurality of second feedforward controls includes an IIR type transfer function in which coefficients are included in a numerator and a denominator, and the controller is configured to adjust, in the first operation, the coefficients included in the numerator and the denominator.
7. The magnetic disk device according to claim 1, wherein the first sensor includes either a vibration sensor or a sensor that detects an amount of the first current supplied to the motor.
8. The magnetic disk device according to claim 2, wherein the plurality of second sensors includes a vibration sensor and a sensor that detects an amount of the first current supplied to the motor.
9. The magnetic disk device according to claim 3, wherein the plurality of second sensors includes a vibration sensor and a sensor that detects an amount of the first current supplied to the motor.
10. The magnetic disk device according to claim 4, wherein the plurality of second sensors includes a vibration sensor and a sensor that detects an amount of the first current supplied to the motor.
11. The magnetic disk device according to claim 1, wherein the controller is configured to monitor a number of processes per unit time of the magnetic disk device, and the set event includes a case where the number of processes is smaller than a third threshold.
12. The magnetic disk device according to claim 2, wherein the controller is configured to monitor a number of processes per unit time of the magnetic disk device, and the set event includes a case where the number of processes is smaller than a third threshold.
13. The magnetic disk device according to claim 3, wherein the controller is configured to monitor a number of processes per unit time of the magnetic disk device, and the set event includes a case where the number of processes is smaller than a third threshold.
14. The magnetic disk device according to claim 4, wherein the controller is configured to monitor a number of processes per unit time of the magnetic disk device, and the set event includes a case where the number of processes is smaller than a third threshold.
15. A method of controlling a magnetic disk device, the magnetic disk device including: a magnetic disk; a magnetic head; an actuator arm having the magnetic head at a distal end; a motor configured to move the actuator arm to move the magnetic head in a radial direction of the magnetic disk; a drive circuit configured to generate a first current that drives the motor by an amount according to an instruction value and supply the generated first current to the motor; and a first sensor configured to detect a predetermined physical quantity, the method comprising: detecting a position of the magnetic head; generating a first instruction value so as to reduce a deviation amount of a detection position of the magnetic head from a target position of the magnetic head; correcting the first instruction value by a first feedforward control based on a first detection value obtained by the first sensor; inputting a second instruction value, which is a value obtained by correcting the first instruction value, to the drive circuit as the instruction value; and executing a first operation in accordance with occurrence of a set event, the first operation being an operation of acquiring a first waveform that is a waveform of the deviation amount, acquiring from the first waveform a first frequency band that is a frequency band whose amplitude is greater than a first threshold, and adjusting a coefficient of a transfer function of the first feedforward control so as to suppress the first detection value in a second frequency band different from the first frequency band and output a correction amount for the first instruction value based on the first detection value in the first frequency band.

16. The method according to claim 15, wherein the magnetic disk device further includes a plurality of second sensors each of which detects a predetermined physical quantity, the plurality of second sensors including the first sensor, and the method further comprises: correcting the first instruction value by a plurality of second feedforward controls, the plurality of second feedforward controls including the first feedforward control, and each of the plurality of second feedforward controls being a control of generating a correction amount for the first instruction value based on a second detection value obtained by a mutually different one of the plurality of second sensors; and adjusting, in the first operation, a coefficient of a transfer function of each of one or more second feedforward controls among the plurality of second feedforward controls so as to suppress the second detection value in the second frequency band and generate a correction amount for the first instruction value based on the second detection value in the first frequency band.

17. The method according to claim 16, further comprising: in the first operation, acquiring a plurality of second waveforms, the plurality of second waveforms being waveforms of second detection values respectively obtained from the plurality of second sensors; acquiring coherence in the first frequency band between each of the plurality of second waveforms and the first waveform; specifying, among the second detection values and the second waveforms, a second feedforward control based on a second detection value constituting a second waveform whose coherence is greater than a second threshold among the plurality of second feedforward controls; and adjusting a coefficient of a transfer function of the specified second feedforward control.

18. The method according to claim 17, wherein each of the plurality of second waveforms includes a waveform during a seek operation, and the first waveform includes a waveform during a seek settling operation.

19. The method according to claim 15, wherein the transfer function includes an IIR type transfer function in which coefficients are included in a numerator and a denominator, and the method further comprises adjusting, in the first operation, the coefficient included in the numerator of the transfer function and the coefficient included in the denominator of the transfer function.

20. The method according to claim 16, wherein the transfer function of each of the plurality of second feedforward controls includes an IIR type transfer function in which coefficients are included in a numerator and a denominator, and the method further comprises adjusting, in the first operation, the coefficients included in the numerator and the denominator.
